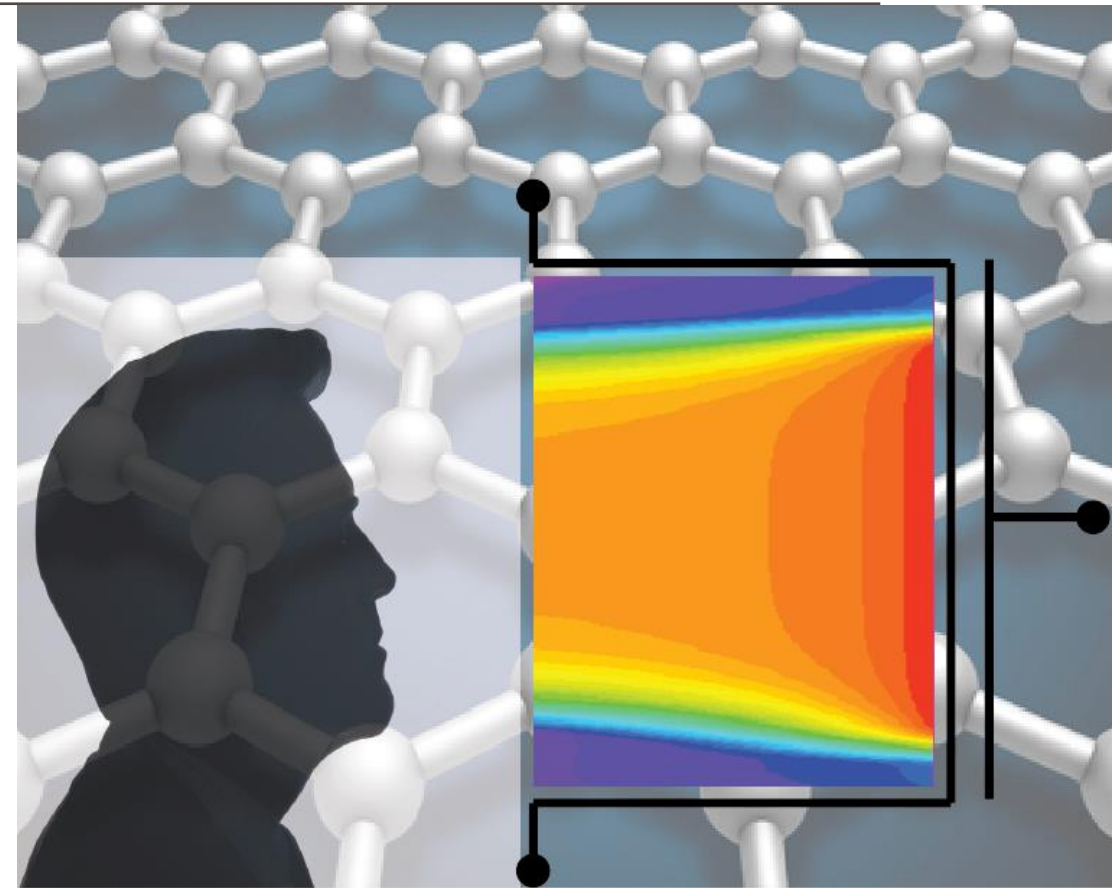


ADVANCED SOLID-STATE DEVICES

Week-5, Lecture-1

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3rd February, 2025



PN Junction: At Equilibrium

PN Junction: Built-in Potential Barrier (2)

Considering hole concentration on the p-side:

$$p_0 = N_a = n_i \exp \left[\frac{E_{Fi} - E_F}{kT} \right]$$

$$p_0 = N_a = n_i \exp \left[\frac{q\phi_{Fp}}{kT} \right]$$

$$\phi_{Fp} = \frac{kT}{q} \ln \left(\frac{N_a}{n_i} \right)$$

Similarly, considering electron concentration on the n-side

$$\phi_{Fn} = -\frac{kT}{q} \ln \left(\frac{N_d}{n_i} \right)$$

Built-in potential is:

$$V_{bi} = |\phi_{Fp}| + |\phi_{Fn}| = \frac{kT}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

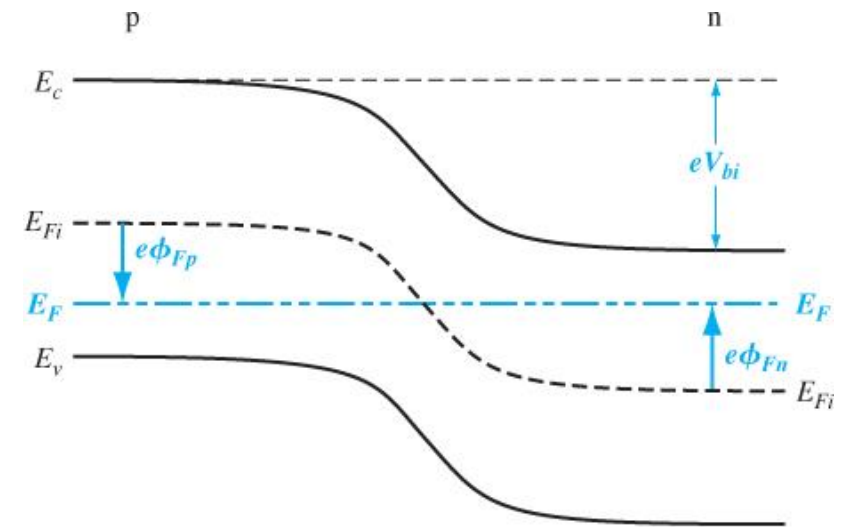


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.

- We refer to $\frac{kT}{q}$ as thermal voltage and denote as V_t

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

PN Junction: Charge Density

Problem: Show the charge density profile in the depletion region of a pn junction assuming that the space charge region abruptly ends at $x = +x_n$ and $x = -x_p$

[x_p is assumed to be positive and $x = 0$ is the boundary of two regions]

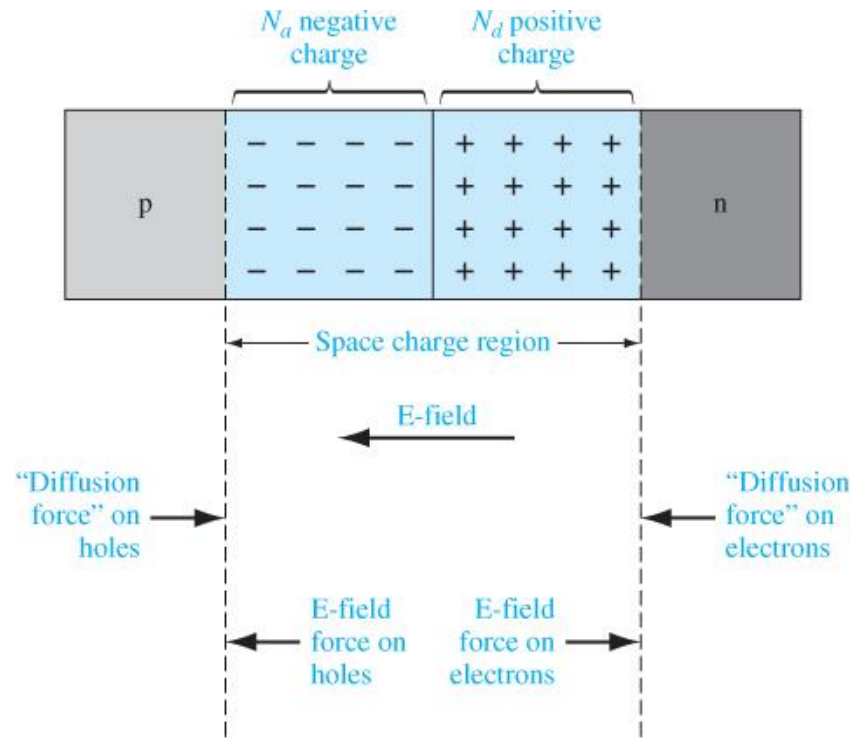


Figure 7.2 | The space charge region, the electric field, and the forces acting on the charged carriers.

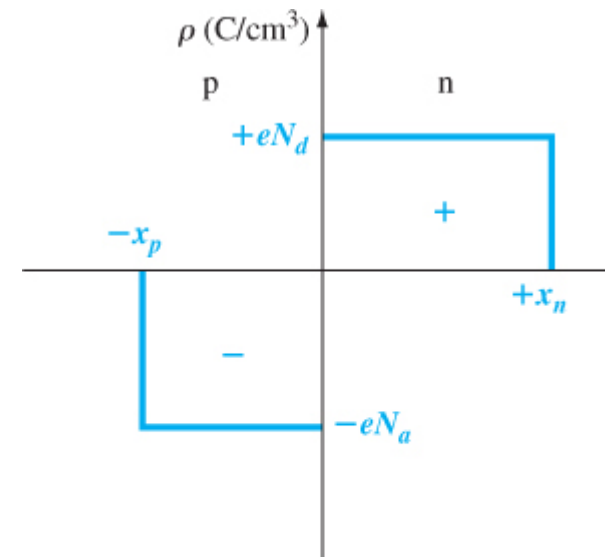


Figure 7.4 | The space charge density in a uniformly doped pn junction assuming the abrupt junction approximation.

PN Junction: Electric Field (1)

Problem: Compute the electric field inside the depletion region of a pn junction. Assume that the electric field is zero outside the depletion region. Assume that the charge density profile is as shown below.

Poisson's Equation:

$$\frac{dE}{dx} = -\frac{d^2\phi}{dx^2} = \frac{\rho}{\epsilon_s}$$

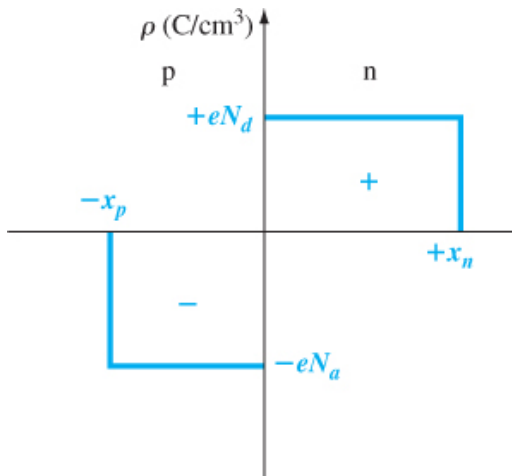


Figure 7.4 | The space charge density in a uniformly doped pn junction assuming the abrupt junction approximation.

Do yourself: Integrate Poisson's equation

PN Junction: Electric Field (2)

$$\text{p-side: } E = \frac{-qN_a}{\epsilon_s}(x + x_p) \text{ for } -x_p \leq x \leq 0$$

$$\text{n-side: } E = -\frac{qN_d}{\epsilon_s}(x_n - x) \text{ for } x_n \geq x \geq 0$$

At $x = 0$, both equations should give same electric field:

$$\text{On the p-side: } E = \frac{-qN_ax_p}{\epsilon_s} \text{ for } x = 0$$

$$\text{On the n-side: } E = \frac{-qN_dx_n}{\epsilon_s} \text{ for } x = 0$$

So x_p and x_n should satisfy:

$$N_ax_p = N_dx_n$$

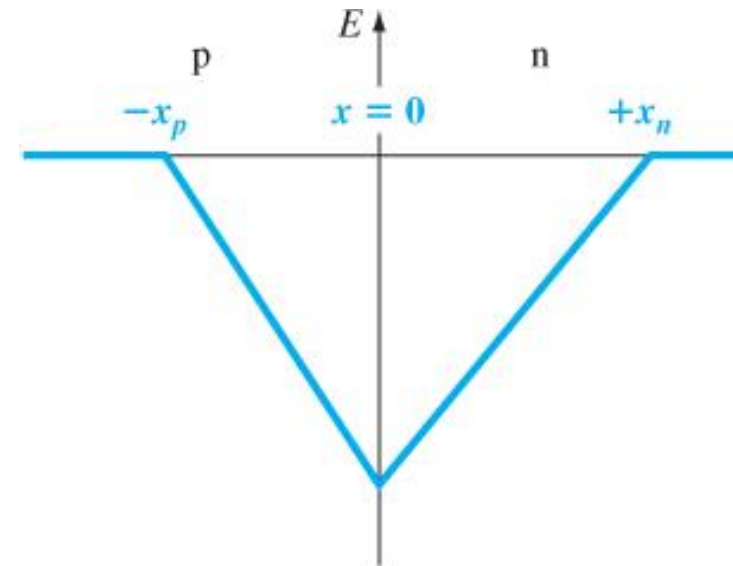


Figure 7.5 | Electric field in the space charge region of a uniformly doped pn junction.

PN Junction: Impact of Built-in Electric Field

What are the consequences of the internal electric field on the carriers in the two regions?

Majority carriers:

- Diffusion of majority carriers blocked by the electric field
 - Holes on p-side blocked
 - Electrons on n-side blocked

Minority carriers:

- Drift of minority carriers supported by the electric field:
 - Electrons on the p-side at the edge of the depletion layer moves towards the junction
 - Holes on the n-side at the edge of the depletion layer moves towards the junction

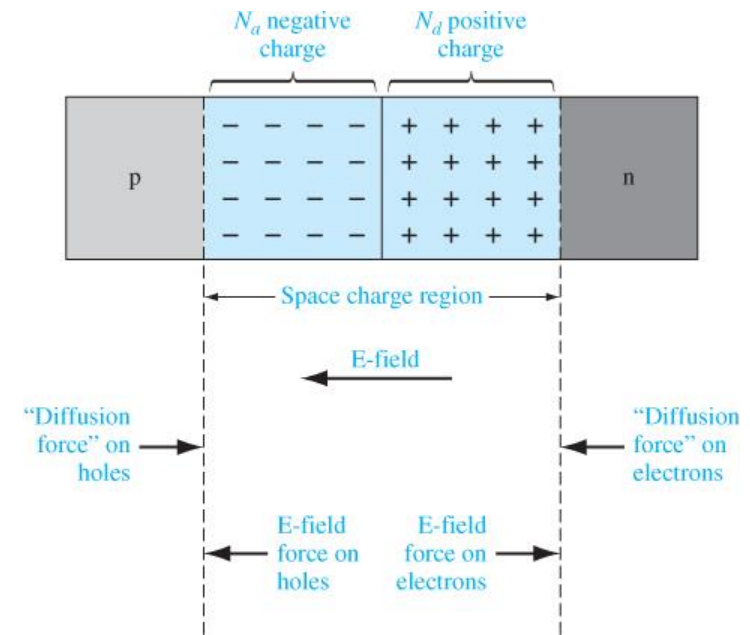


Figure 7.2 | The space charge region, the electric field, and the forces acting on the charged carriers.

At equilibrium drift and diffusion balance each other for both electrons and holes

PN Junction: Doping and Depletion Layer Width

x_p and x_n should satisfy: $N_a x_p = N_d x_n$

What are its implications?

$$x_p = \frac{N_d}{N_a} x_n$$

- If doping on p-side is less, the extent of depletion layer width on the p-side will be more
- If doping on n-side is less, the extent of depletion layer width on the n-side will be more

- **Problem:** If a junction is p^+n , on which side will be the major part of the depletion layer?
- Such pn junctions are called *one-sided junctions*

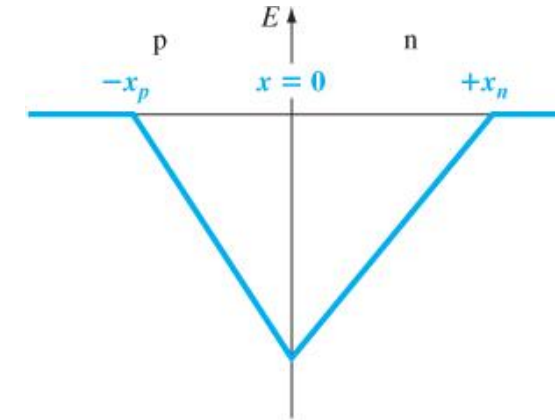


Figure 7.5 | Electric field in the space charge region of a uniformly doped pn junction.

PN Junction: Electric Potential (1)

Problem: Compute the electric potential inside the depletion region of a pn junction. Assume that the electric potential is zero at $x = -x_p$. Assume that the electric field is as shown below. Assume $\phi(x) = 0$ at $x = -x_p$

$$E = -\frac{d\phi}{dx}$$

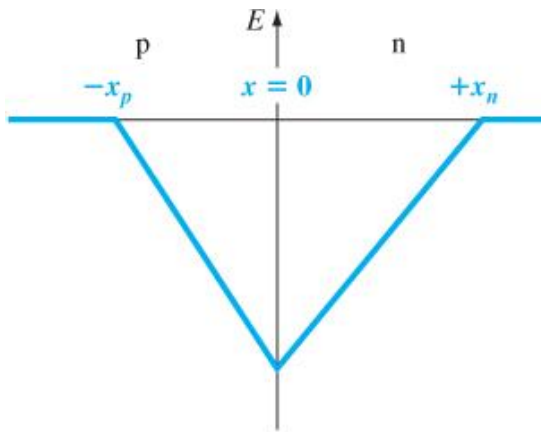


Figure 7.5 | Electric field in the space charge region of a uniformly doped pn junction.

Do yourself

$$\text{Integrating: } \int_0^{\phi(x)} d\phi = \int_{-x_p}^x \frac{qN_a}{\epsilon_s} (x + x_p) dx$$

$$\phi(x) = \frac{qN_a(x + x_p)^2}{2\epsilon_s}$$

$$\text{p-side: } E = \frac{-qN_a}{\epsilon_s} (x + x_p)$$

PN Junction: Electric Potential (2)

On the p-side:

$$\phi(x) = \frac{qN_a(x + x_p)^2}{2\epsilon_s}$$

At $x = 0$, $\phi(x) = \frac{qN_ax_p^2}{2\epsilon_s}$

On the n-side: $E = -\frac{qN_d}{\epsilon_s}(x_n - x)$

On the n-side (On integration):

$$\phi(x) = \frac{qN_d}{\epsilon_s} \left[x_n x - \frac{x^2}{2} \right] + \frac{qN_ax_p^2}{2\epsilon_s}$$

$$\phi(x_n) = \frac{q}{2\epsilon_s} (N_dx_n^2 + N_ax_p^2)$$

$\phi(x_n)$ is same as built-in potential

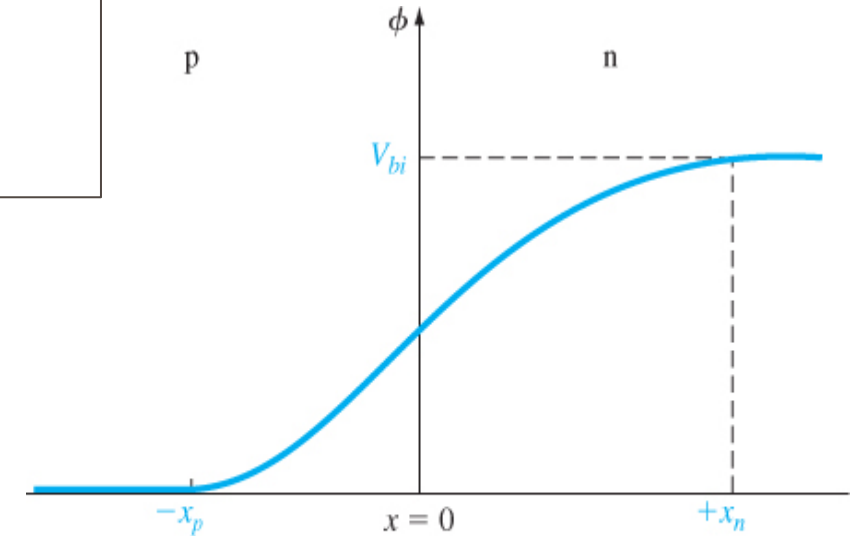


Figure 7.6 | Electric potential through the space charge region of a uniformly doped pn junction.

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

- The built-in potential is decided by the doping
- The extent of the depletion region (x_p and x_n) adjusts accordingly

PN Junction: Width of Depletion Region (1)

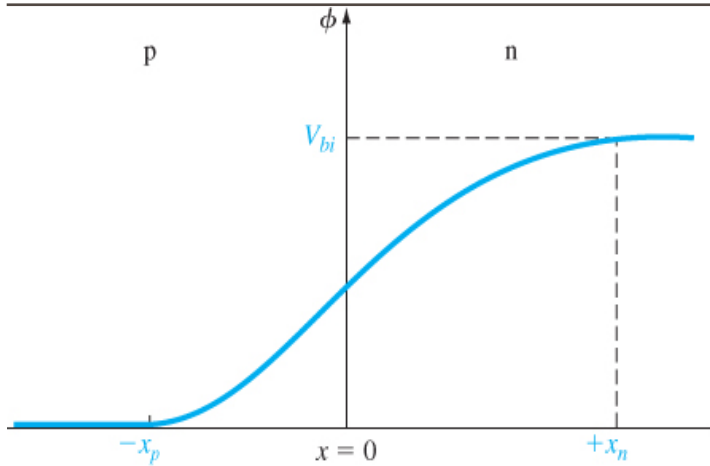


Figure 7.6 | Electric potential through the space charge region of a uniformly doped pn junction.

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

Similarly, we can compute

$$x_p = \sqrt{\frac{2\epsilon_s V_{bi}}{q(N_a + N_d)} \left[\frac{N_d}{N_a} \right]}$$

$$V_{bi} = \frac{q}{2\epsilon_s} (N_d x_n^2 + N_a x_p^2)$$

x_p and x_n should satisfy:

$$N_a x_p = N_d x_n$$

Problem: Using above expressions, compute x_n, x_p in terms of V_{bi}

Do yourself:

$$x_n = \sqrt{\frac{2\epsilon_s V_{bi}}{q(N_a + N_d)} \left[\frac{N_a}{N_d} \right]}$$

$$W = x_n + x_p = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left[\frac{(N_a + N_d)}{N_a N_d} \right]}$$

PN Junction in Equilibrium: Problem

Consider a PN junction at equilibrium at 300 K with $N_a = 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$

Determine the built-in potential and the width of the depletion layer.

[Take $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon_s = 11.7 \times 8.85 \times 10^{-14} \text{ F/cm}$]

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

$$V_{bi} = 0.0259 \times \ln \left(\frac{10^{16} \times 10^{15}}{2.25 \times 10^{20}} \right) \\ = 0.635 \text{ V}$$

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left[\frac{(N_a + N_d)}{N_a N_d} \right]}$$

$$W = \sqrt{\frac{2 \times 11.7 \times 8.85 \times 10^{-14} \times 0.635}{1.6 \times 10^{-19}} \left[\frac{1.1 \times 10^{16}}{10^{31}} \right]} \\ = 9.51 \times 10^{-5} \text{ cm} = 0.951 \text{ } \mu\text{m}$$

PN Junction: Depletion Approximation (1)

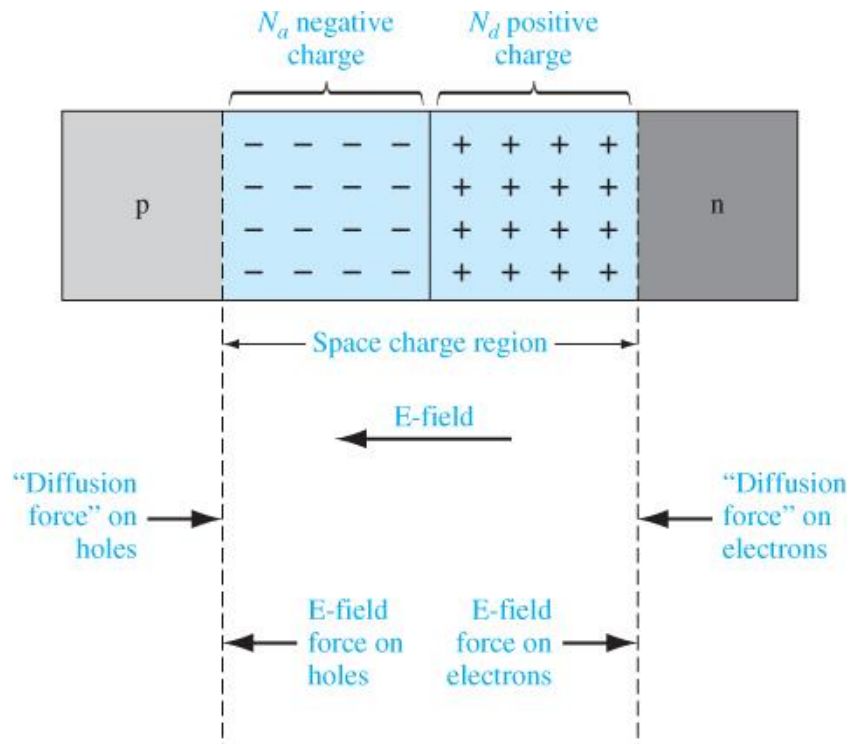


Figure 7.2 | The space charge region, the electric field, and the forces acting on the charged carriers.

What assumptions have we made?

We have made following assumptions in the earlier analysis of pn junctions:

1. Transition from neutral region to space charge region (depletion region) is abrupt
2. There are no mobile carriers within the space charge (depletion) region

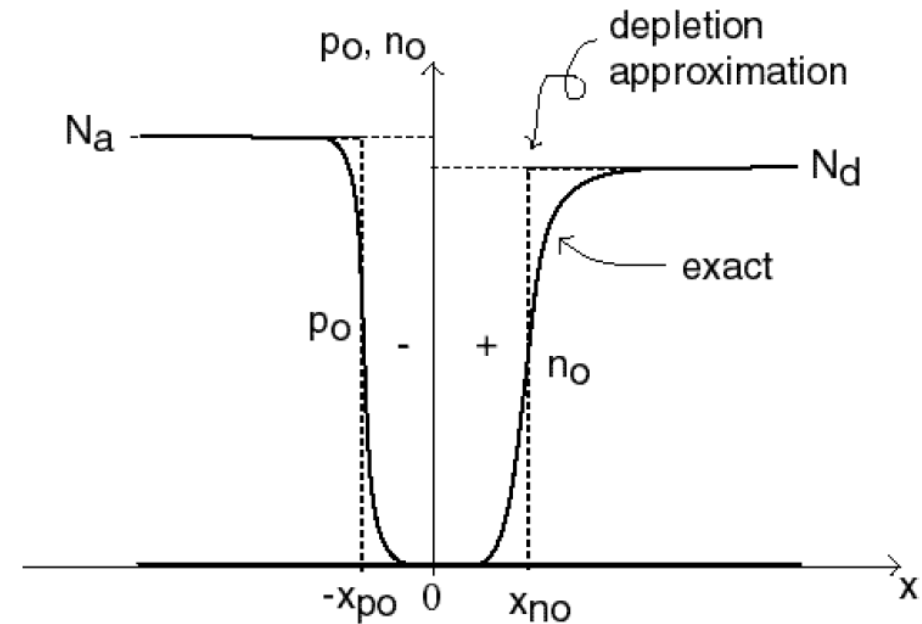
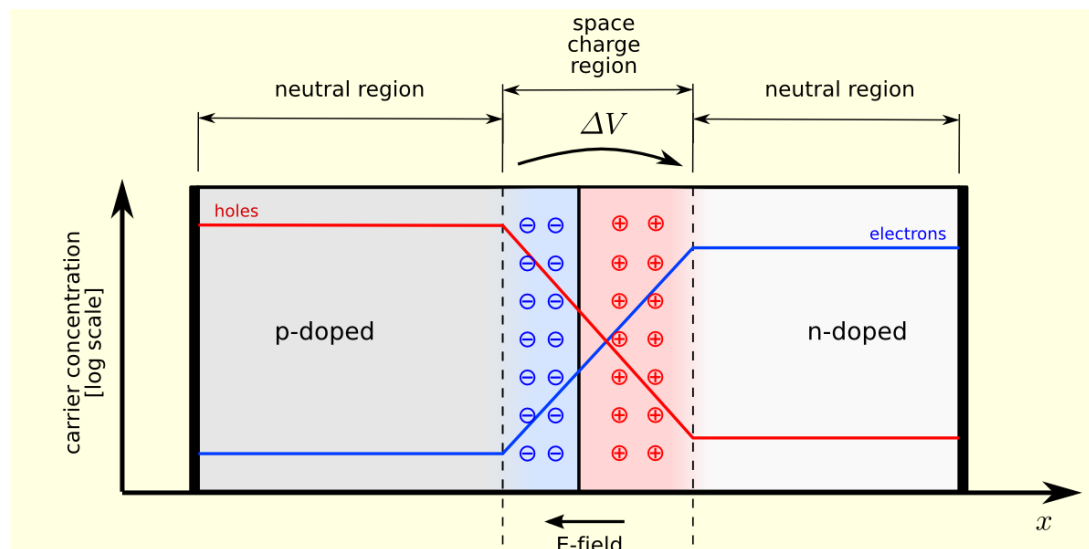
- These approximations are known as *depletion approximation*

- We should be careful of the depletion approximation

PN Junction: Depletion Approximation (2)

What are deviations from the depletion approximation?

1. The edges are not sharp
2. The mobile carriers decrease exponentially in the depletion region

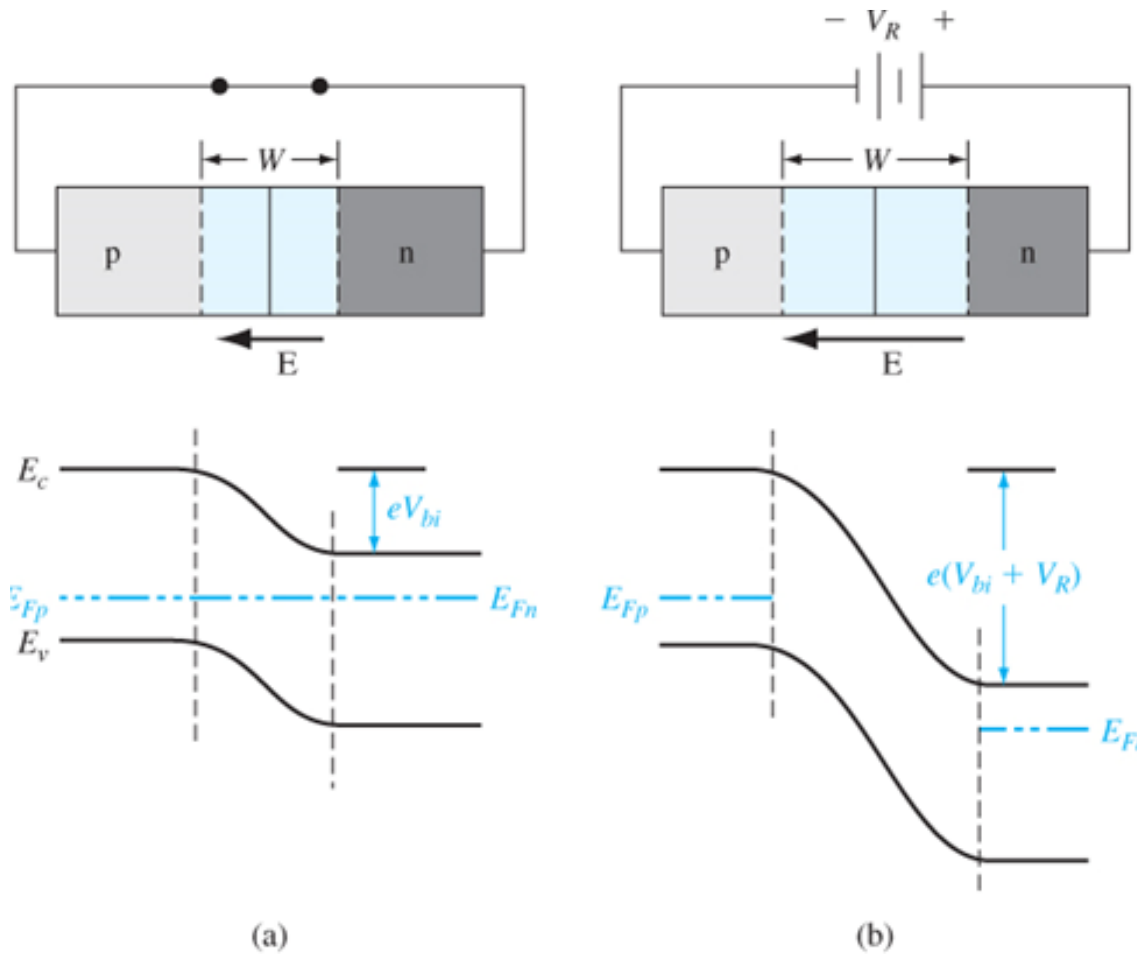


PN Junction



Applying Bias

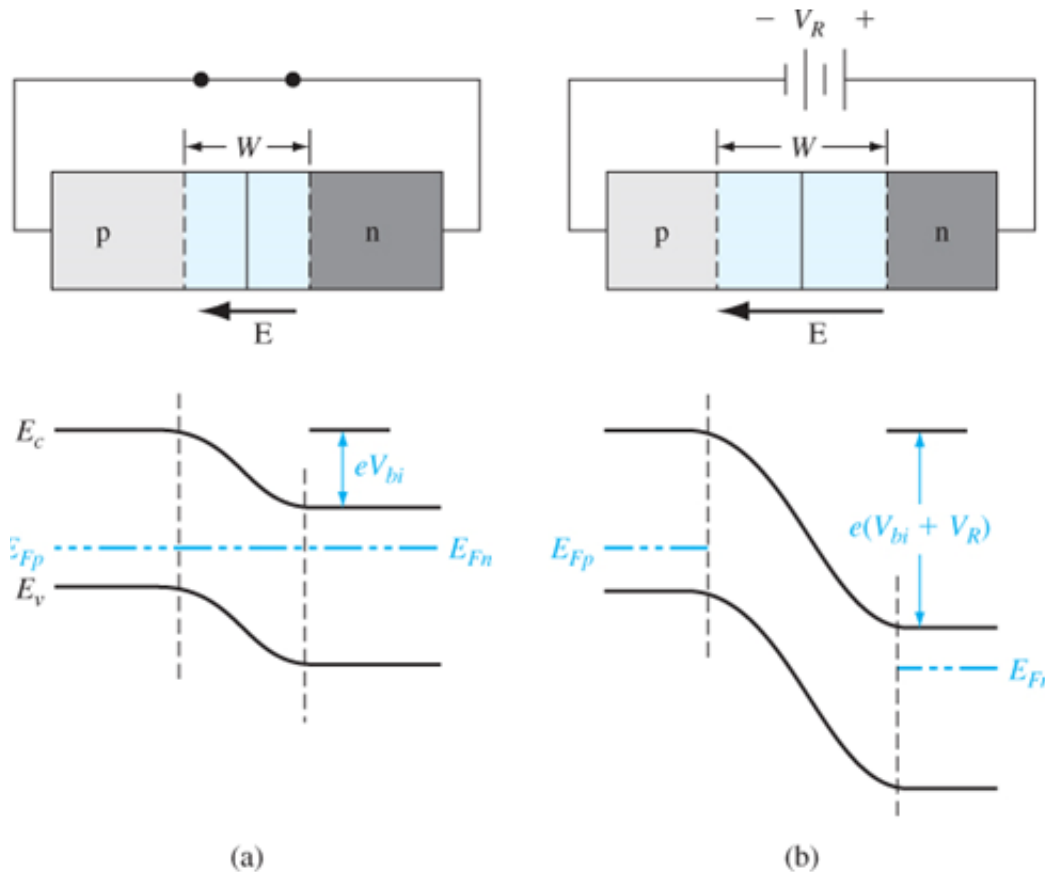
PN Junction: Reverse Bias (1)



Energy Band View:

- Positive to n-side and negative to p-side
- Band edges move downward on n-side
 - because positive voltage reduces energy of electrons
- Barrier seen by the electron increase

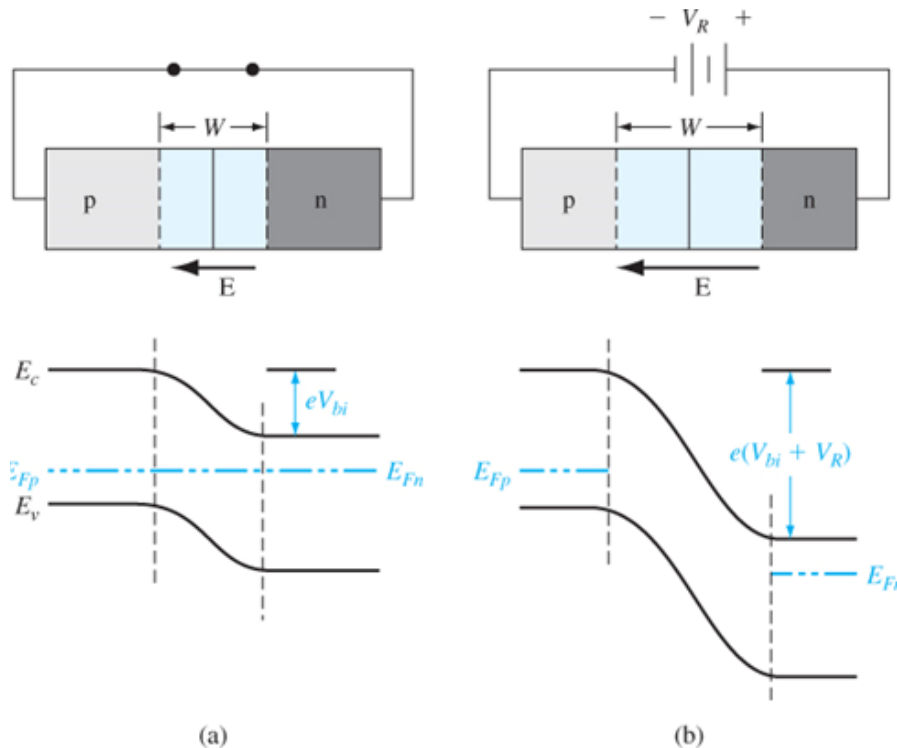
PN Junction: Reverse Bias (2)



Charge View:

- Positive voltage on the electrons on n-side pulls them out from the edge of the depletion region
- Negative voltage on holes on p-side pulls them out from edge of depletion region
- As a result, depletion region width increase (potential barrier increases)

PN Junction: Reverse Bias Current

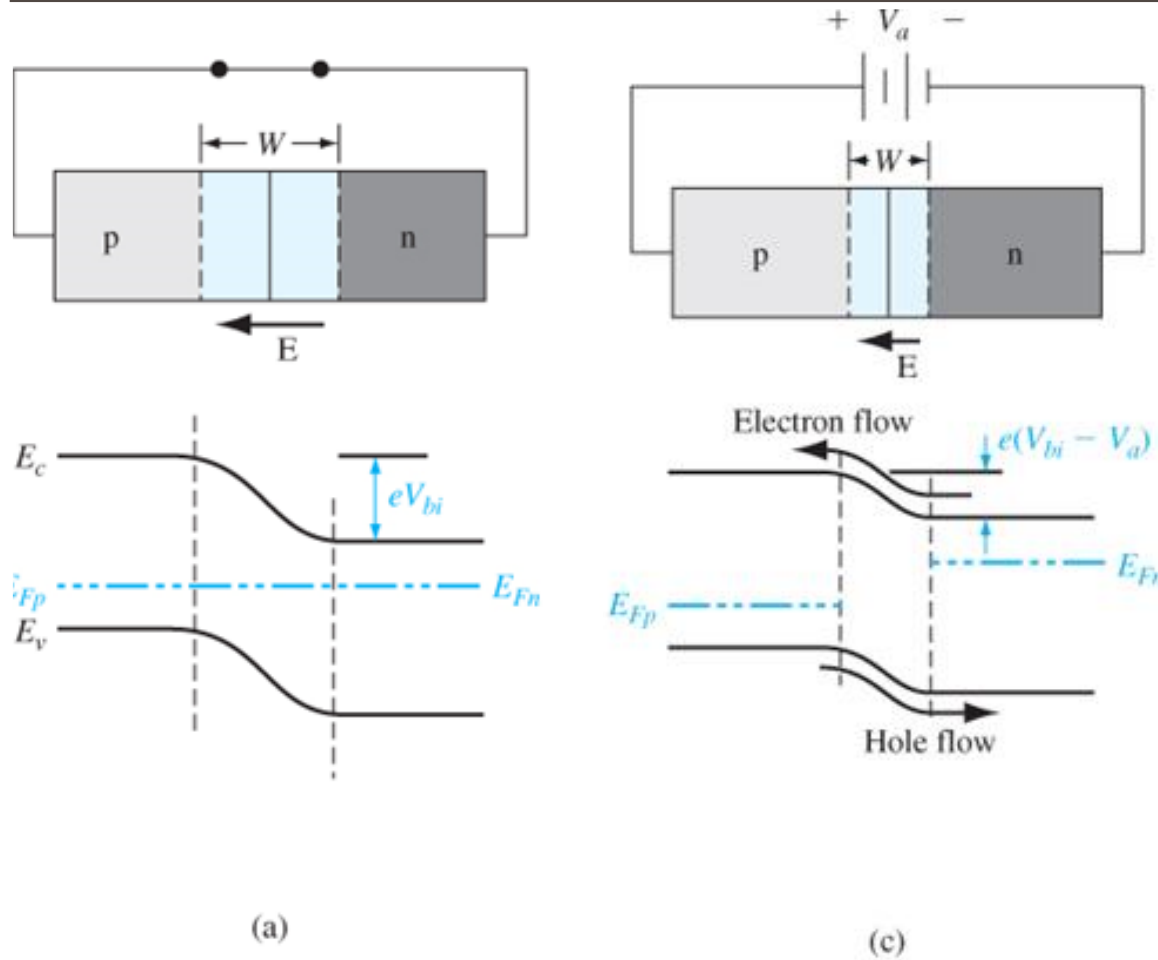


Current due to minority carriers (small)

Due to increase in the potential barrier:

- Less electrons can diffuse over the barrier from the n-side to the p-side
- Less holes can diffuse over the barrier from the p-side to the n-side
- The drift component of minority carriers remains practically same
 - Some electrons (minority carrier) of p-side can reach to the boundary of the depletion region on the n-side and constitute reverse current due to drift
 - Some holes (minority carrier) of the n-side can reach to the boundary of the depletion region on the p-side and constitute reverse current due to drift

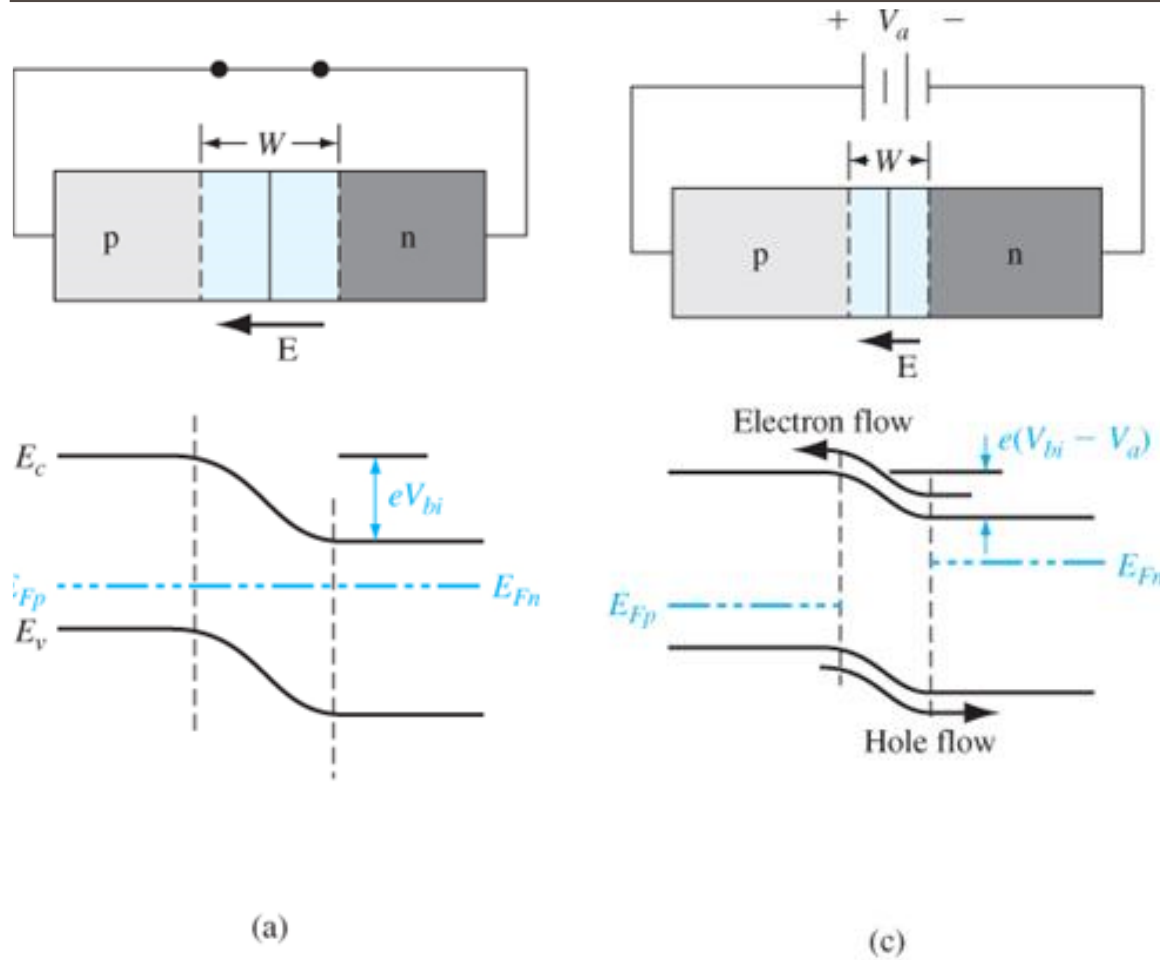
PN Junction: Forward Bias (1)



Energy Band View:

- Positive to p-side and negative to n-side
- Band edges move downward on p-side
 - because positive voltage reduces energy of electrons
- Barrier seen by the electron decreases

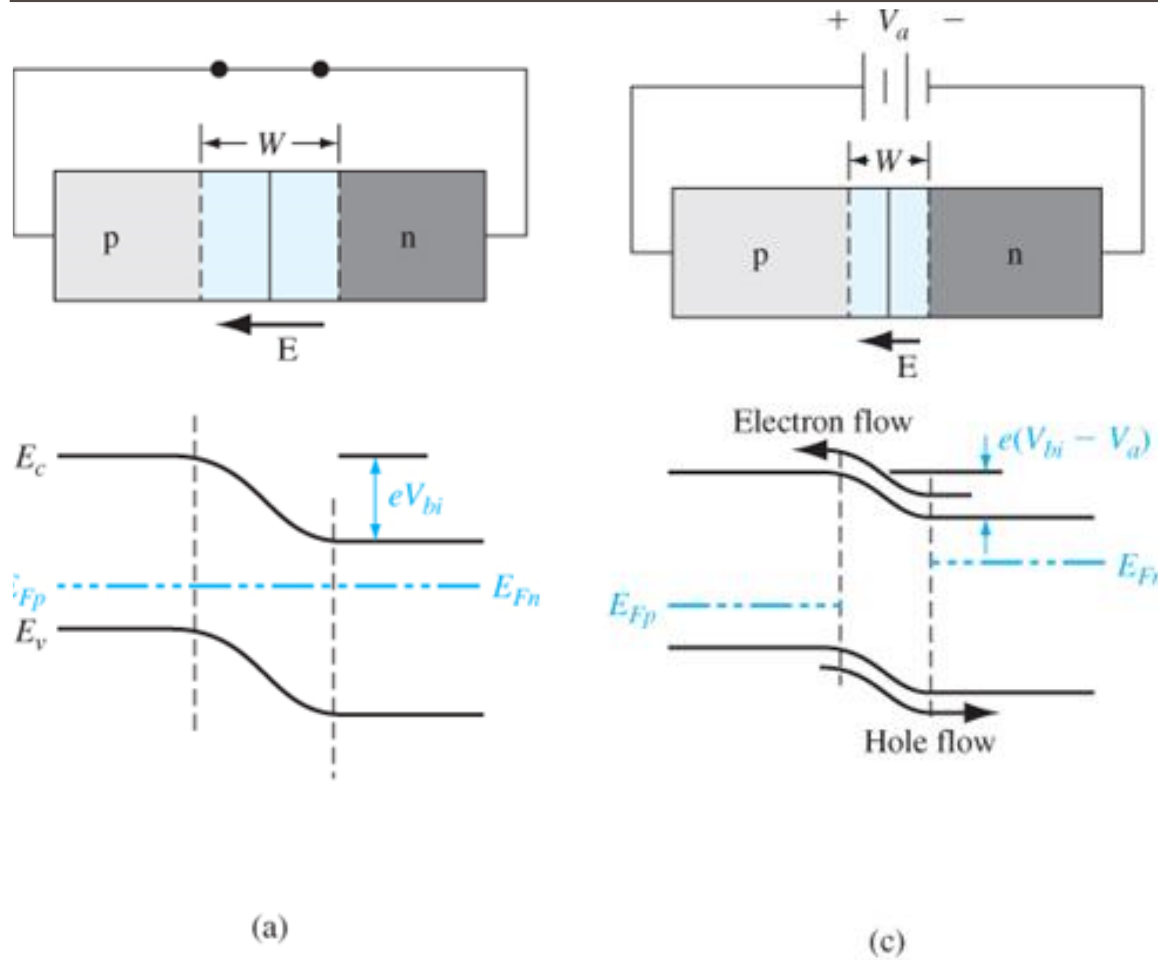
PN Junction: Forward Bias (2)



Charge View:

- Positive voltage on the p-side pushes holes towards depletion region
- Negative voltage on electrons on n-side pushes them towards depletion region
- As a result, charges neutralizes in the depletion region and its width decreases (potential barrier decreases)

PN Junction: Forward Bias Current



Due to reduction in the potential barrier:

- More electrons can diffuse over the barrier from the n-side to the p-side
- More holes can diffuse over the barrier from the p-side to the n-side
- The drift component of minority carriers remains practically same

Current due to majority carriers (high)

PN Junction



Reverse Bias

Reverse Bias: Electric Field

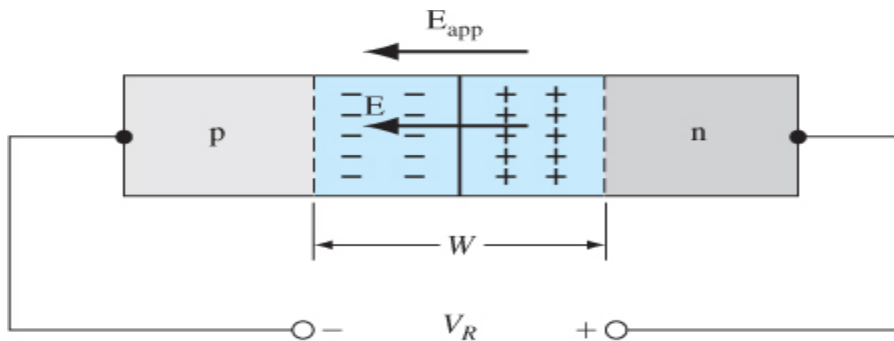


Figure 7.8 | A pn junction, with an applied reverse-biased voltage, showing the directions of the electric field induced by V_R and the space charge electric field.

- Applied electric field and internal electric field are in same direction (Total electric field increases)

- The peak electric field increases

- Peak electric field at junction:

$$E_{max} = \frac{-qN_a x_p}{\epsilon_s} = \frac{-qN_d x_n}{\epsilon_s}$$

- x_n and x_p increases due to reverse bias

Substitute total potential barrier for built-in potential :

$$E_{max} = -\sqrt{\frac{2q(V_{bi} + V_R)}{\epsilon_s} \left[\frac{N_a N_d}{(N_a + N_d)} \right]}$$

Reverse Bias: Junction Capacitance

$$x_n = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{q(N_a + N_d)} \left[\frac{N_a}{N_d} \right]}$$

$$x_p = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{q(N_a + N_d)} \left[\frac{N_d}{N_a} \right]}$$

- Depletion region width:

$$W = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{q} \left[\frac{(N_a + N_d)}{N_a N_d} \right]}$$

- Charge is stored in the depletion region

- Charge per unit area: in the depletion region

$$Q_d = qN_a x_p = qN_d x_n$$

- The magnitude of charge changes with applied bias

➤ because x_n and x_p depends on V_R

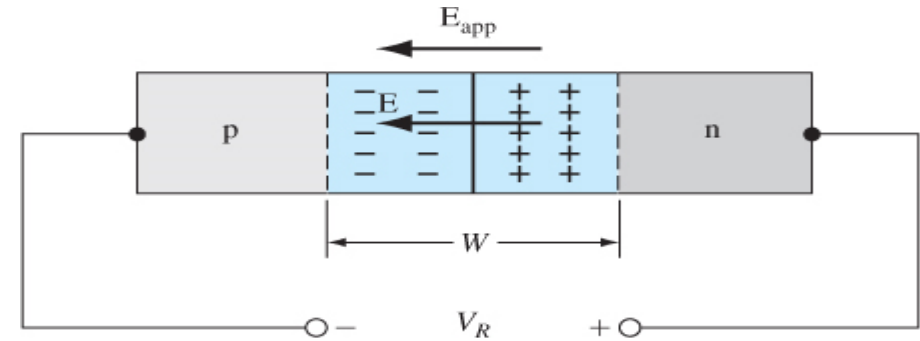


Figure 7.8 | A pn junction, with an applied reverse-biased voltage, showing the directions of the electric field induced by V_R and the space charge electric field.

- The change in x_n and x_p is made by majority carriers moving in/out of the depletion region
- The situation is similar to a parallel plate capacitor
 - $Q = CV$
 - $dQ = CdV$ increase in voltage lead to increase in charge stored]

Junction Capacitance vs. Parallel Plate Capacitor

What are differences between Junction Capacitor and parallel plate capacitor?

Junction Capacitor	Parallel Plate Capacitor
Charge is distributed throughout the space.	Charge resides on two plates
Charge changes non-linearly with voltage (because width changes non-linearly with voltage) $Q = f(V)$, where f is a non-linear function	Charge changes linearly with voltage $Q = CV$

- We want to capture this non-linear relationship using a single quantity (capacitance) similar to a parallel plate capacitor
- How can we do it?
 - Linearize it at a given voltage

- We define junction capacitance mathematically as:

$$C_j = \left. \frac{dQ}{dV} \right|_{V_R}$$

- Junction Capacitance is dependent on applied bias

Junction Capacitance: Mathematically

Problem: Compute the junction capacitance for a reverse-biased pn junction. The junction capacitance is defined as:

$$C_j = \left. \frac{dQ}{dV} \right|_{V_R}$$

$$C_j = \frac{dQ}{dV_R} = \frac{dQ}{dx_n} \frac{dx_n}{dV_R} = qN_d \frac{dx_n}{dV_R}$$

$$\frac{dx_n}{dV_R} = \frac{1}{2} \left(\frac{2\epsilon_s(V_{bi} + V_R)}{q(N_a + N_d)} \left[\frac{N_a}{N_d} \right] \right)^{-0.5} \frac{2\epsilon_s}{q(N_a + N_d)} \left[\frac{N_a}{N_d} \right]$$

$$= \sqrt{\frac{\epsilon_s}{2q(N_a + N_d)(V_{bi} + V_R)} \left[\frac{N_a}{N_d} \right]}$$

Charge per unit area:

$$Q = qN_a x_p = qN_d x_n$$

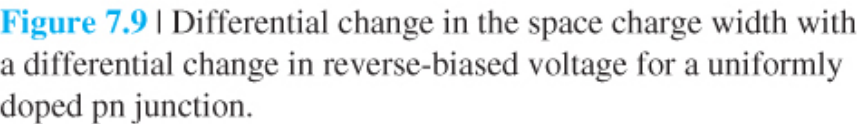
$$x_n = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{q(N_a + N_d)} \left[\frac{N_a}{N_d} \right]}$$

Capacitance per unit area

$$C_j = \sqrt{\frac{q\epsilon_s N_a N_d}{2(N_a + N_d)(V_{bi} + V_R)}}$$

$$W = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{q} \left[\frac{(N_a + N_d)}{N_a N_d} \right]}$$

$$C_j = \frac{\epsilon_s}{W}$$



- DC applied voltage: V_R
- Incremental voltage applied dV_R
- As a result, increase in charge dQ

- The difference in charge is separated by depletion layer width (similar to a parallel plate capacitor)
- We can write capacitance per unit area:
$$C_j = \frac{\epsilon_s}{W}$$

- *Junction capacitance* is also known as *depletion layer capacitance*.
- It depends on the reverse bias and built-in potential

PN Junction in Equilibrium: Problem

Consider a PN junction at equilibrium at 300 K with $N_a = 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$

Determine the junction capacitance at $V_R = 5 \text{ V}$.

[Take $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon_s = 11.7 \times 8.85 \times 10^{-14} \text{ F/cm}$]

$$V_{bi} = 0.0259 \times \ln \left(\frac{10^{16} \times 10^{15}}{2.25 \times 10^{20}} \right) \\ = 0.635 \text{ V}$$

$$C_j = \sqrt{\frac{q\epsilon_s N_a N_d}{2(N_a + N_d)(V_{bi} + V_R)}}$$

$$C_j = \sqrt{\frac{1.6 \times 10^{-19} \times 11.7 \times 8.85 \times 10^{-14} \times 10^{31}}{2 \times (1.1 \times 10^{16})(0.635 + 5)}} \\ = 3.66 \times 10^{-9} \text{ F/cm}^2$$

If area is $A = 10^{-4} \text{ cm}^2$, capacitance will be 0.366 pF.

References

- D. A. Neaman, "*Semiconductor Physics and Devices*"